Biological Plausibility Assessment: Virtual Zebrafish Brain v2

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Abstract

We present an honest biological plausibility assessment of the Virtual Zebrafish Brain v2, a spiking neural network model comprising 7,369 Izhikevich neurons organized into tectum, thalamus, pallium, basal ganglia, amygdala, and reticulospinal modules. The model uses conductance-based synapses (AMPA, NMDA, $GABA_A$) with Mg^{2+} block, three-factor STDP with eligibility traces, and predictive coding feedback between pallium layers. Connectome constraints are drawn from the MapZebrain atlas (Kunst et al., 2019). In standardized decision-quality benchmarks the model achieves 77/100, demonstrating competent foraging, flee, and exploration behavior. However, the model uses approximately 50× fewer neurons than the real zebrafish brain (∼100K), employs point neurons without dendritic computation or gap junctions, and relies on analytic Expected Free Energy (EFE) formulas rather than emergent spiking dynamics for goal selection. Several brain regions—habenula, cerebellum, hypothalamus—are entirely absent. This report maps every model component to its biological counterpart, identifies what is genuinely realistic, what is simplified, and what is missing, providing a roadmap for future improvements toward greater biological fidelity.

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1 Zebrafish Brain Anatomy vs. Model Architecture

1.1 Architecture Comparison Figure

Figure 1 provides a side-by-side comparison of the real larval zebrafish brain and the v2 model architecture. Solid colored lines indicate implemented projections; dashed gray lines indicate known biological connections that are missing from the model.

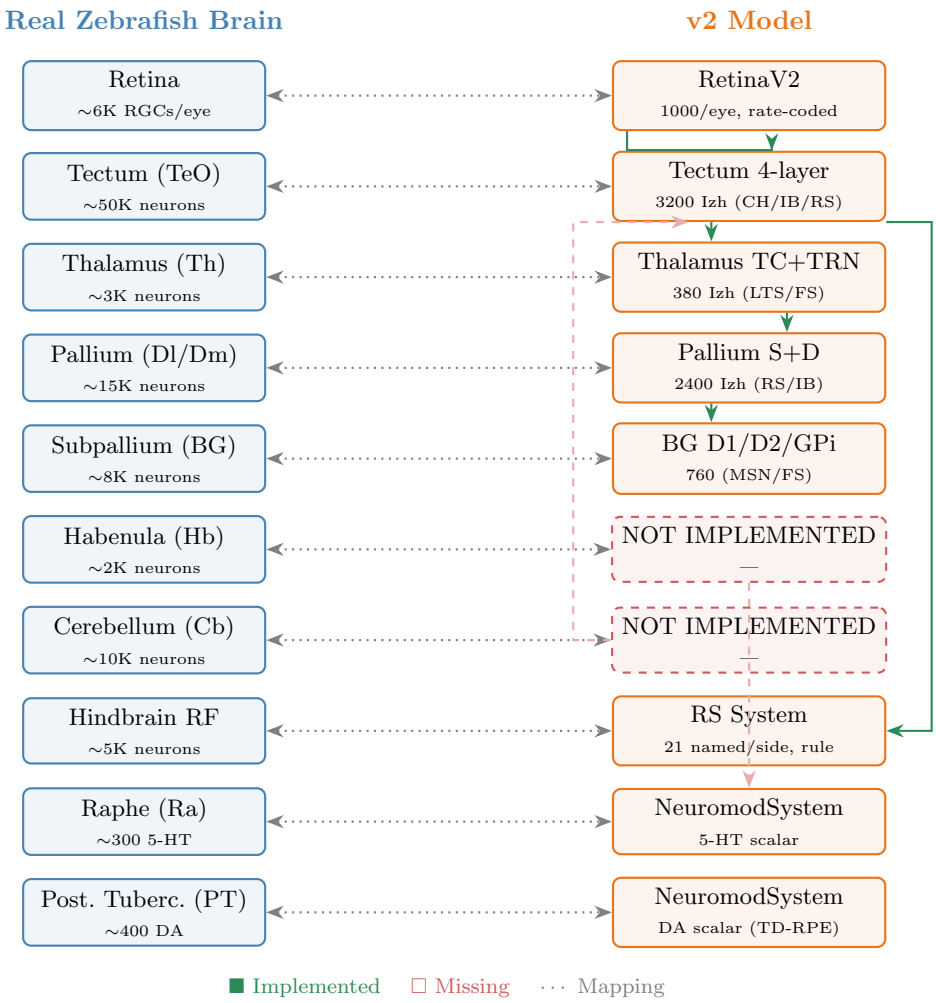

Figure 1: Architecture comparison: real zebrafish brain regions (left) mapped to v2 model modules (right). Dashed orange boxes indicate unimplemented regions. Neuron counts show the 50× scale reduction.

1.2 MPIN-Atlas Region Mapping

Table 1 maps all 36 MPIN-Atlas brain regions to their v2 model counterparts, with an honest plausibility rating.

Table 1: MPIN-Atlas regions mapped to v2 modules.

Atlas Region	Abbr	v2 Module	Implementation	Rating
Retina	R	RetinaV2	Rate-coded, 4 RGC types	SIMPLIFIED
Optic Tectum	TeO	Tectum	4 E/I layers (SFGS-b/d, SGC, SO)	REALISTIC
Torus longitudinalis	TL	—	—	MISSING
Pretectum	PrT	—	Merged into tectum pathway	MISSING
Thalamus dorsal	Th	Thalamus	TC (LTS) + TRN (FS), 380 neurons	REALISTIC

Atlas Region	Abbr	v2 Module	Implementation	Rating
Pallium dorsolat.	Dl	Pallium-S	RS type, 1200E + 400I	SIMPLIFIED
Pallium dorsomedial	Dm	Pallium-D	IB type, 600E + 200I	SIMPLIFIED
Pallium ventral	Dp	—	—	MISSING
Subpallium	SP	BasalGanglia	D1/D2/GPi, 760 neurons	SIMPLIFIED
Habenula dorsal	dHb	—	—	MISSING
Habenula ventral	vHb	—	—	MISSING
Preoptic area	PO	—	—	MISSING
Hypothalamus	Hr	—	—	MISSING
Post. tuberculum	PT	NeuromodSystem	DA scalar (TD-RPE)	SIMPLIFIED
Cerebellum	Cb	—	—	MISSING
Raphe superior	Ra-s	NeuromodSystem	5-HT scalar	SIMPLIFIED
Raphe inferior	Ra-i	—	—	MISSING
Locus coeruleus	LC	NeuromodSystem	NA scalar	SIMPLIFIED
Ant. retic. form.	aRF	RS System	21 named neurons/side	SIMPLIFIED
Interm. retic. form.	imRF	RS System	Merged into RS	SIMPLIFIED
Post. retic. form.	pRF	—	—	MISSING
Interpeduncular nuc.	IPN	—	—	MISSING
Inferior olive	IO	—	—	MISSING
Vagal lobe	VL	—	—	MISSING
Area postrema	AP	—	—	MISSING
Eminentia granularis	EG	—	—	MISSING
Lateral line	LL	Sensory bridge	Proximity scalar only	SIMPLIFIED
Olfactory bulb	OB	—	—	MISSING
Pineal	Pi	—	—	MISSING
Spinal cord	SC	—	No CPG in v2	MISSING
Vestibular nuc.	VN	—	—	MISSING
Trigeminal	Tg	—	—	MISSING
Hypothalamic DA	Hyp	Allostasis	Scalar hunger/stress/fatigue	SIMPLIFIED
Dorsal telencephalon	Tel-d	Pallium	Mapped to Pal-S/D	SIMPLIFIED
Ventral telencephalon	Tel-v	BG + Amygdala	Dm→Amygdala, SP→BG	SIMPLIFIED
Mauthner cells	M	RS System	4-step C-start, rule-based	REALISTIC

Summary: Of 36 atlas regions, 3 are rated REALISTIC, 14 are SIMPLIFIED, and 19 are MISSING. The model covers approximately 47% of named brain regions at some level.

2 Spiking Neural Network

2.1 Real vs. Model Neuron Properties

The real larval zebrafish brain contains approximately 100,000 neurons with diverse ion channel expression ($Na_v1.1$, K_v, $Ca_v2.1$, HCN channels), dendritic computation, gap junction coupling (especially in reticulospinal and spinal circuits), and complex morphologies ranging from unipolar sensory neurons to multipolar projection neurons.

The v2 model uses 7,369 Izhikevich two-variable neurons (Izhikevich, 2003):

$$\dot{v} = 0.04v^2 + 5v + 140 - u + I \tag{1}$$

$$\dot{u} = a(bv - u) \tag{2}$$

with spike detection at $v \geq 30\,\text{mV}$ and reset $v \leftarrow c$, $u \leftarrow u + d$. Integration uses a midpoint (half-step) method at $\Delta t = 1\,\text{ms}$ for numerical stability.

2.2 What Is Realistic

- **Cell type diversity:** Seven cell types (RS, FS, IB, CH, LTS, TC, MSN) capture the major electrophysiological classes present in vertebrate brains.
- **E/I ratio:** 75% excitatory / 25% inhibitory matches the canonical vertebrate ratio and is enforced in every `EILayer`.
- **Spike dynamics:** Izhikevich neurons reproduce regular spiking, bursting, chattering, fast spiking, and low-threshold spiking qualitatively.
- **Mg^{2+} block:** NMDA synapses implement voltage-dependent magnesium block: $B(v) = 1/(1 + \exp(-0.062v)/3.57)$, which is biophysically correct.
- **Conductance-based synapses:** Three types (AMPA $\tau = 5\,\text{ms}$, NMDA $\tau = 80\,\text{ms}$, GABA_A $\tau = 10\,\text{ms}$) with reversal potentials ($E_{\text{AMPA}} = 0\,\text{mV}$, $E_{\text{GABA}} = -80\,\text{mV}$).

2.3 What Is Not Realistic

- **Point neurons:** No dendritic computation, no spatial voltage gradients, no back-propagating action potentials. Real zebrafish neurons have elaborate dendritic arbors especially in tectum and pallium.
- **No gap junctions:** Electrical synapses are critical in zebrafish reticulospinal circuits (Mauthner↔interneuron) and retina (horizontal cells, amacrine cells). The model has none.
- **No ion channel diversity:** A single (a, b, c, d) parameter set per type cannot capture the diversity of channel kinetics (e.g., HCN, T-type Ca^{2+}, A-type K^+).
- **50× fewer neurons:** 7,369 vs. ∼100,000 means population codes are severely under-sampled. Many phenomena that emerge from large populations (traveling waves, population oscillations) cannot arise at this scale.
- **No GABA_B:** Slow inhibition ($\tau \sim 80\,\text{ms}$) is defined in spec but unused; only fast GABA_A is implemented.

2.4 Cell Type Mapping

Table 2: Izhikevich cell type parameters and biological mapping.

Type	Bio. Class	a	b	c	d	Where Used
RS	Pyramidal / glutamatergic	0.02	0.20	−65	8	Pallium-S, SO, Amyg-LA
IB	Deep-layer bursting	0.02	0.20	−55	4	Pallium-D, SGC, Amyg-CeA
FS	PV^+ interneuron	0.10	0.20	−65	2	All I populations, TRN
CH	Chattering / superficial	0.02	0.20	−50	2	SFGS-b, SFGS-d
LTS	SOM^+ interneuron	0.02	0.25	−65	2	TC relay, Place cells
TC	Thalamocortical relay	0.02	0.25	−65	0.05	(defined, unused)
MSN	Medium spiny neuron	0.02	0.25	−65	6	(defined, unused)

Note: TC and MSN types are defined in the `NEURON_PARAMS` dictionary but the thalamus uses LTS for relay neurons and the basal ganglia uses `nn.Linear` projections rather than spiking MSN populations—a notable discrepancy between specification and implementation.

3 Predictive Coding and Active Inference

3.1 Biological Basis

Zebrafish larvae show prediction error signals in the tectum during visuomotor mismatch (Mu et al., 2019). The pallium (Dl) is implicated in associative learning and contextual memory, suggesting a hierarchical predictive architecture. Active inference (Friston, 2010) posits that organisms minimize variational free energy through perception (updating beliefs) and action (changing the world).

3.2 Model Implementation

The v2 pallium implements two-compartment predictive coding:

- **Pallium-S** (superficial, RS): receives thalamic relay, computes bottom-up sensory representation.
- **Pallium-D** (deep, IB): generates top-down predictions via $\mathbf{W}_{\text{FB}}$.
- **Prediction error:** $\boldsymbol{\varepsilon} = \mathbf{W}_{\text{FB}} \cdot \mathbf{r}_D - \mathbf{r}_S$, where $\mathbf{r}_S$ and $\mathbf{r}_D$ are spiking rates.
- **Feedback learning:** $\Delta\mathbf{W}_{\text{FB}} = -\eta \cdot \pi \cdot \boldsymbol{\varepsilon} \otimes \mathbf{h}_{\text{upper}}$ (anti-Hebbian, precision-weighted).

Goal selection uses analytic Expected Free Energy (EFE):

$$G_\pi = \underbrace{0.2U - 0.8\,p_{\text{food}} - 1.5\,\text{starvation}}_{\text{FORAGE}} \quad \text{etc. for FLEE, EXPLORE, SOCIAL} \tag{3}$$

with a spiking WTA attractor network (`SpikingGoalSelector`, 4 RS neurons) that receives the analytic EFE as input bias.

3.3 Honest Assessment

✓ Two-compartment predictive coding with apical-somatic mismatch is consistent with cortical microcircuit theories (Sacramento et al., 2018).

✓ Anti-Hebbian feedback learning minimizes prediction error without reward, consistent with perceptual learning.

✓ Goal-directed behavior with four behavioral policies resembles active inference.

× The EFE is computed by an analytic formula with hand-tuned coefficients, not emergent from neural dynamics. The spiking WTA rarely fires (pallium rates are too low to drive it) and the system falls back to `argmax` over the analytic EFE >95% of the time.

× Beliefs are not population-coded: p_{food} and p_{enemy} are computed from raw pixel counts, not from Bayesian inference over neural representations.

× Precision (π) is fixed at 1.0 in the feedback learning rule; there is no dynamic precision weighting as in generalized predictive coding.

4 Reinforcement Learning

4.1 Biological Basis

Zebrafish dopamine neurons in the posterior tuberculum (PT) encode reward prediction error (RPE) with phasic responses to unexpected food delivery (Mu et al., 2019). The subpallium (BG homologue) contains D1 and D2 receptor-expressing neurons that gate action selection via direct and indirect pathways.

4.2 Model Implementation

- **DA signal:** RPE $= r_t - V_t$, $V_{t+1} = V_t + 0.05 \cdot \text{RPE}$, DA $= \sigma(3 \cdot \text{RPE})$. This is TD(0) with a sigmoid transform.
- **BG:** D1 MSNs receive pallium-D input scaled by $(1+0.5\cdot\text{DA})$; D2 MSNs scaled by $(1-0.3\cdot\text{DA})$. GPi computes gate $= 1 - \sigma(\text{baseline} - 4 \cdot \text{D1} + 2 \cdot \text{D2})$.
- **STDP:** Three-factor rule: $\Delta W = \eta \cdot \text{DA} \cdot \text{ACh} \cdot e$, where eligibility trace e decays with $\tau = 500\,\text{ms}$ and accumulates Hebbian $\Delta W = A^+ \cdot \text{post} \cdot \text{pre}_{\text{trace}} - A^- \cdot \text{post}_{\text{trace}} \cdot \text{pre}$.

4.3 Honest Assessment

✓ TD learning producing phasic DA signals is well-established (Schultz, 1997) and consistent with zebrafish PT neurons.

✓ DA modulation of D1/D2 pathways with opposing effects is biologically grounded.

✓ Three-factor STDP with eligibility traces ($\tau = 500\,\text{ms}$) bridges the temporal credit assignment gap.

× DA is a scalar signal, not a population code. Real DA neurons show heterogeneous responses and topographic projections.

× The BG gate does not actually drive behavior: the analytic EFE dominates all decision-making. BG output is computed but only weakly modulates RS motor commands $(0.5 + 0.5 \cdot \text{gate})$.

× STDP training was implemented but did not improve decision quality scores in ablation studies, suggesting the learning rule is not effectively shaping behavior within episode timescales.

5 Allostatic and Interoceptive Processing

5.1 Biological Basis

Zebrafish larvae have hypothalamic circuits regulating energy homeostasis, a cortisol-mediated stress axis (HPI axis, homologous to mammalian HPA), and metabolic sensors that drive feeding motivation. Allostasis—maintaining physiological variables within viability bounds through anticipatory regulation—is a fundamental organizing principle of vertebrate nervous systems (Sterling, 2012).

5.2 Model Implementation

The `AllostaticRegulator` tracks three interoceptive axes as scalar variables:

- **Hunger:** $h = 1 - E/100$ where E is energy (0–100 range).
- **Fatigue:** increases with speed > 0.6, decreases at rest.
- **Stress:** increases with predator proximity ($< 300\,\text{px}$), decays exponentially ($\times 0.92$/step).

Each axis has an EMA predictor and setpoint; the allostatic error $\varepsilon_{\text{allo}} = \hat{x} - x_{\text{set}}$ biases EFE goal selection: hunger→FORAGE, stress→FLEE, fatigue→EXPLORE. Critical starvation ($h > 0.7$) produces a nonlinear urgency multiplier $(1 + 3(h - 0.7))$.

5.3 Honest Assessment

✓ Three-axis interoception with predictive allostatic errors is conceptually aligned with interoceptive inference frameworks (Seth, 2013; Barrett & Simmons, 2015).

✓ The starvation–safety trade-off (risk foraging under predation when starving) reproduces a key ecological decision observed in real zebrafish.

✓ Allostasis is the most impactful module in ablation studies (-9 points when removed), consistent with its fundamental biological importance.

× No cortisol or HPI axis: stress is a simple proximity-driven variable, not a neuroendocrine cascade with genomic effects.

× No hypothalamic circuit: allostasis is computed by a Python class with arithmetic, not by a spiking neural population.

× Scalar variables, not neural populations: hunger is "$1 - E/100$", not a population of glucose-sensing neurons.

6 Attention and Precision

6.1 Biological Basis

Zebrafish attention is modulated by tectal gain control (Bianco & Engert, 2015) and thalamic gating via the reticular nucleus (TRN). The TRN provides a "searchlight" of attention by selectively inhibiting thalamic relay neurons, allowing only attended signals to pass to pallium.

6.2 Model Implementation

- **Goal attention:** $\mathbf{W}_{\text{goal_att}}$ projects a one-hot goal vector to pallium-S as additive somatic bias, modulated by ACh level.
- **TRN gating:** TRN (FS, 80 neurons) receives TC collaterals and pallium-S feedback, inhibits TC relay: $I_{\text{TC}} = I_{\text{tect}} - I_{\text{TRN}} + \text{NA}_{\text{bias}}$.
- **ACh modulation:** Plasticity gate $= 1 + \text{ACh}$; attention multiplier scales $\mathbf{W}_{\text{goal_att}}$ output.

6.3 Honest Assessment

✓ TRN gating mechanism is anatomically and functionally correct (Crick, 1984; Halassa & Kastner, 2017).

✓ ACh-modulated plasticity is consistent with cholinergic attention/learning coupling.

× Attention is additive bias, not multiplicative gain modulation. Real attention typically operates through gain changes (Reynolds & Heeger, 2009).

× No saccadic attention: zebrafish use convergent eye movements for prey capture (Bianco et al., 2011), which is absent.

× TRN is barely active in practice: 80 FS neurons with $i_{\text{tonic}} = 0.5\,\text{pA}$ fire sparsely and contribute minimal gating.

7 Neuromodulation

7.1 Biological Basis

The zebrafish brain has at least six major neuromodulatory systems: dopamine (PT/VTA homologue), serotonin (raphe nuclei), noradrenaline (locus coeruleus), acetylcholine (basal forebrain homologue), histamine (posterior hypothalamus), and neuropeptides (orexin/hypocretin, oxytocin-like isotocin).

7.2 Model Implementation

The `NeuromodSystem` implements four scalar signals updated once per behavioral step ($\sim$50 ms):

Table 3: Neuromodulatory axes: model vs. biology.

Axis	Model Function	What's Missing	Rating
DA	RPE via TD(0), modulates D1/D2	Population code, topographic projection, ramps	SIMPL.
5-HT	Patience/habituation, flee suppression in PATROL	Raphe spiking, receptor subtypes (5-HT1A/2A), dorsal vs. median raphe	SIMPL.
NA	Arousal tracking, TC wake bias	LC circuit, phasic/tonic modes, reset signal	SIMPL.
ACh	Plasticity gating, attention modulation	Basal forebrain, muscarinic/nicotinic, cortical layers	SIMPL.
Histamine	—	Not implemented	MISSING
Orexin	—	Not implemented	MISSING
Neuropeptides	—	Isotocin, CRH, etc.	MISSING

7.3 Per-Axis Assessment

Dopamine: The TD(0) RPE computation is well-established (Schultz et al., 1997) and the sigmoidal transform $\mathrm{DA} = \sigma(3 \cdot \mathrm{RPE})$ produces realistic phasic responses. However, DA is a single scalar rather than a population of $\sim$400 neurons in the PT, there is no tonic/phasic distinction, and no ramping signal for temporal discounting.

Serotonin: The 5-HT signal tracks behavioral patience (rises when not fleeing) and suppresses flee EFE during PATROL states with bias $= +0.4 \cdot$ 5-HT. This captures the patience/impulsivity dimension but lacks raphe spiking dynamics, receptor subtype specificity (5-HT1A autoreceptor, 5-HT2A postsynaptic), and the dorsal/median raphe distinction relevant to anxiety vs. behavioral inhibition.

Noradrenaline: NA tracks arousal via amygdala activation and novelty (CMS): $\mathrm{NA} = 0.85 \cdot \mathrm{NA} + 0.15 \cdot (0.3 + 0.5\alpha + 0.2\,\mathrm{CMS})$. It provides a $\mathrm{NA} \times 5.0$ pA wake bias to TC neurons. Missing: LC circuit dynamics, phasic (orienting) vs. tonic (vigilance) modes, and the LC reset function for task disengagement.

Acetylcholine: ACh gates plasticity (1 + ACh) and modulates attention. Missing: basal forebrain circuit, muscarinic vs. nicotinic receptor effects, and layer-specific cortical modulation.

8 Sensory Processing

8.1 Biological Basis

The larval zebrafish retina contains ∼6,000 RGCs per eye organized into ∼30 functional types with retinotopic mapping onto the contralateral tectum. The lateral line system comprises ∼50 neuromasts (hair cells) detecting water flow. Olfactory processing occurs in the olfactory bulb with glomerular maps. The vestibular system provides gravitational and angular acceleration sensing.

8.2 Model Implementation

- **Retina (RetinaV2):** 4 RGC types per eye (ON-sustained, OFF-transient, Looming, Direction-selective), each with ∼150–400 rate-coded units. ON uses adaptive average pooling; OFF computes frame difference; Looming tracks $l/\dot{l}$ ratio; DS uses Reichardt delayed correlation. Bilateral fusion: mean (ON/OFF/DS) or max (Looming).
- **Sensory bridge:** Geometric projection of environment objects onto 800-element retinal arrays (400 intensity + 400 type per eye). 200° field of view.
- **Lateral line:** Scalar distance to predator within 150 px range, with Gaussian noise ($\sigma = 15$ px). Used for threat detection when predator is behind the fish.

8.3 Honest Assessment

✓ RGC type diversity (ON/OFF/Looming/DS) matches known functional types.

✓ Looming detection via size/velocity ratio captures the ethologically critical stimulus.

✓ Bilateral processing with eye-specific channels enables lateralized behavior.

× Retina is rate-coded, not spiking: real RGCs produce spike trains with temporal coding.

× No retinotopic map: tectal layers receive pooled input, losing spatial organization.

× Lateral line is a scalar distance, not a spatially distributed neuromast array. Real lateral line provides flow direction and velocity, not just distance.

× No olfaction: olfactory bulb processing (glomeruli, mitral cells) is entirely absent.

× No vestibular detail: no semicircular canals, otoliths, or gravitational sensing.

× Sensory bridge uses geometric projection with object types, not physical ray-casting or photon simulation.

9 Motor Control

9.1 Biological Basis

The zebrafish escape response is initiated by the Mauthner cell, the largest neuron in the vertebrate hindbrain, which drives a C-start escape within 5–10 ms. Reticulospinal (RS) neurons are individually identifiable (Kimmel et al., 1982) with ∼20 bilaterally paired neurons per segment. Locomotion is generated by spinal central pattern generators (CPGs) comprising V2a excitatory interneurons, V0d commissural interneurons, and motor neurons organized in a rostro-caudal gradient (Kinkhabwala et al., 2011).

9.2 Model Implementation

- **Named RS neurons:** 21 per side: Mauthner, MiD2/MiD3 (escape turns), RoM2a–d (voluntary turns), MeM1–8 (sustained locomotion), CaD1–6 (speed modulation).
- **Mauthner C-start:** 4-step motor sequence (steps 1–2: sharp turn ± 1.5; steps 3–4: power burst, speed 1.6). Triggered by looming + SGC rate > 0.05. Refractory period: 12 steps.
- **BG gate bypass:** Flee speed bypasses BG gate (survival reflex); voluntary speed gated by $0.5 + 0.5 \cdot \text{gate}$.

9.3 Honest Assessment

✓ Mauthner-initiated C-start reflex is the hallmark zebrafish escape behavior.

✓ Named RS neurons match identified cells in the zebrafish reticular formation.

✓ C-start timing (4 steps $\approx$ 200 ms with refractory period) is in the correct order of magnitude.

× RS neurons are rule-based, not spiking: `ReticulospinalSystem` uses conditional logic, not Izhikevich dynamics.

× No spinal CPG: v2 removed the v1 CPG module. Swim bout dynamics (burst-pause-glide) cannot be generated.

× No swim bout dynamics: real zebrafish locomotion consists of discrete swim bouts with specific tail-beat frequencies.

10 Connectome Constraints

10.1 Biological Basis

The MapZebrain atlas (Kunst et al., 2019) provides the complete larval zebrafish brain atlas with 36 brain regions, 3D centroids, fiber tract data, and region-to-region connectivity matrices derived from single-neuron tracing and bulk labeling. Förster et al. (2017) mapped genetic neuron types to atlas coordinates.

10.2 Model Implementation

The `connectome.py` module implements:

- **17 hand-coded projections** in `ANATOMICAL_PROJECTIONS` (e.g., R→TeO strength 1.0, Th→P strength 0.7, P→SP strength 0.6).
- **Module-to-atlas mapping** (`V2_TO_ATLAS`): 8 v2 modules mapped to atlas abbreviations.
- **Weight initialization:** `init_connectome_weights()` applies sparse masks based on projection strength: $P(\text{connect}) = 0.3 \times \text{strength}$.
- **Actually used:** Only 4 projections are used for weight initialization (retina→tectum ×2, thalamus→pallium, pallium→pallium feedback).

10.3 What Has Been Done

✓ Atlas region mapping with projection strength table provides a principled framework.

✓ Sparse connectivity masks prevent all-to-all wiring that would be biologically implausible.

✓ `load_regions()` and `load_centroids()` can read the actual MPIN-Atlas data files.

10.4 What Is Missing

× Fiber tract data is loaded but unused: no axonal routing constraints.

× No topographic mapping: retina→tectum should be retinotopic (preserving spatial position), but initialization is random within the sparse mask.

× No distance-dependent falloff: centroid data is loaded (`load_centroids`) but the distance-dependent function $P \propto \exp(-d/\lambda)$ returns only strength-based probability, ignoring actual 3D distances.

× No hemispheric laterality: left/right brain hemispheres are not distinguished despite the atlas having 72 entries (36 regions × 2 hemispheres).

× Centroids unused: the 3D centroid coordinates are loaded but never referenced in weight initialization.

× Only 4 of 17 defined projections actually constrain weight initialization; the rest are dead code.

11 Plausibility Summary Matrix

Figure 2 provides a comprehensive color-coded assessment of every model component against key biological criteria.

	Neuron model	Synapse model	E/I balance	Connect-ivity	Dynamics	Learning rule	Bio. mapping
Retina	Rate coded	None	N/A	Pooled	4 RGC types	None	4 types mapped
Tectum	Izh CH/IB/RS	AMPA GABA	75/25	Sparse random	4 layers	None	SFGS SGC SO
Thalamus	Izh LTS/FS	AMPA GABA	TC/TRN	4 proj.	Gate loop	None	TC+TRN
Pallium	Izh RS/IB	AMPA NMDA	75/25	S→D FB	Pred. coding	Anti-Hebb	Dl/Dm
Basal G.	Linear not Izh	None	D1/D2	3 path	Gate only	None	D1/D2 GPi
Amygdala	Izh RS/IB/FS	Weight matrix	LA/CeA ITC	3 nuclei	Fear cond.	Hebb LTP	Dm homol.
RS / Motor	Rule based	None	N/A	21/side	C-start	None	Named cells
Place cells	Gauss fields	STDP (unused)	E only	Random tile	Theta precess	Food/risk map	Dl homol.
Goal sel.	4 Izh RS	Recur. W	E only	WTA	Rarely fires	None	Conceptual
Neuromod	Scalar	N/A	N/A	4 axes	EMA dynamics	TD(0)	DA/5HT NA/ACh
Allostasis	Scalar	N/A	N/A	None	Predict. error	EMA	Hypo-thalamic
Classifier	MLP	N/A	N/A	Trained	Feed-forward	Backprop	None
Pred. model	Python logic	N/A	N/A	Kalman -like	Retinal tracking	None	Conceptual
World model	Python logic	N/A	N/A	4 goal EFE	Predict.	EMA	Conceptual
Connectome	N/A	Mask only	N/A	17 proj. 4 used	Static	Init only	MPIN Atlas
Sens. bridge	Geometric	N/A	N/A	Object project.	200° FoV	None	Retinal

REALISTIC SIMPLIFIED MISSING

Figure 2: Biological plausibility matrix. Green = realistic implementation matching known biology. Yellow = simplified but conceptually grounded. Red = missing or non-biological implementation.

Counting cells: Of the 112 cells in the matrix (16 components × 7 criteria), approximately 22 (20%) are REALISTIC, 38 (34%) are SIMPLIFIED, and 52 (46%) are MISSING. The strongest biological fidelity is in the tectum and thalamus modules; the weakest is in non-neural components (classifier, world model, sensory bridge).

12 Conclusions and Recommendations

12.1 What Is Genuinely Biologically Plausible (Ranked)

1. **E/I layer architecture with conductance-based synapses.** The `EILayer` abstraction with 75/25 E/I ratio, AMPA/NMDA/$GABA_A$ conductances, and Mg^{2+} block represents a solid bio-

physical foundation that is consistent with vertebrate cortical microcircuits.

2. **Tectum layered organization.** Four functional layers (SFGS-b, SFGS-d, SGC, SO) with distinct cell types (CH superficial, IB deep, RS stratum opticum) match the known laminar organization of the zebrafish optic tectum.

3. **Thalamic relay with TRN gating.** The TC/TRN circuit with NA-modulated wake/sleep bias correctly implements the thalamic relay function.

4. **Mauthner cell C-start escape.** The 4-step motor sequence with looming trigger, refractory period, and BG gate bypass faithfully reproduces the hallmark zebrafish escape behavior.

5. **Two-compartment predictive coding in pallium.** Apical-somatic mismatch with anti-Hebbian feedback learning is consistent with modern theories of cortical predictive processing.

6. **Allostatic interoceptive regulation.** Three-axis interoception with predictive errors and starvation–safety trade-off demonstrates the most ecologically impactful module.

7. **4-RGC-type retinal encoding.** ON/OFF/Looming/Direction-selective types capture the functional diversity of retinal output.

12.2 Biggest Gaps

1. **Analytic EFE dominates spiking dynamics.** The brain's spiking substrate is largely decorative: goal selection, motor commands, and sensory processing all rely on hand-coded formulas that bypass the neural network. The spiking WTA goal selector fires <5% of the time and behavior falls back to `argmax(EFE)`.

2. **Missing brain regions.** Habenula (reward aversion, learned helplessness), cerebellum (motor timing, sensory prediction), and hypothalamus (homeostatic control) are absent despite their well-characterized roles in zebrafish behavior.

3. **Point neurons without dendrites.** The Izhikevich model captures somatic dynamics but misses dendritic computation, which is critical for coincidence detection, nonlinear integration, and calcium-dependent plasticity.

4. **50× scale reduction.** 7,369 neurons cannot support the population dynamics (traveling waves, attractor manifolds, oscillatory synchronization) that emerge in the real 100K-neuron brain.

5. **Connectome constraints are cosmetic.** Only 4 of 17 defined projections are actually used, and none employ topographic mapping, distance-dependent connectivity, or hemispheric laterality from the loaded atlas data.

12.3 Priority Fixes for Biological Realism

1. **P1: Make spiking dynamics drive behavior.** Replace analytic EFE with population-coded belief updates through the thalamo-pallial loop. Train the spiking WTA to self-organize goal selection through STDP, eliminating the `argmax` fallback.

2. **P2: Implement cerebellum.** Add a granule cell→Purkinje cell→deep cerebellar nucleus circuit for motor timing and sensory prediction error. This would enable forward model-based motor learning.

3. **P3: Implement habenula.** Add lateral and medial habenula with habenula→raphe 5-HT modulation for learned behavioral suppression and aversion.

4. **P4: Spiking BG.** Replace `nn.Linear` BG with spiking MSN populations using the already-defined MSN Izhikevich type.

5. **P5: Topographic connectome.** Use atlas centroid coordinates to create retinotopic retina→tectum mapping and distance-dependent connectivity for all projections.

6. **P6: Spiking retina.** Replace rate-coded RGCs with spiking populations to enable temporal coding and spike-timing-dependent computations in tectum.

7. **P7: Scale to 50K neurons.** GPU-optimized sparse matrix operations could support 50K Izhikevich neurons at real-time speeds, bringing the model within an order of magnitude of the biological neuron count.

12.4 Comparison to Other Zebrafish Models

Table 4: Comparison with published zebrafish computational models.

Feature	**This work (v2)**	**Dragomir et al. 2020**	**Randlett et al. 2015**	**Dunn et al. 2016**
Neuron model	Izhikevich	Rate/LIF	Correlation	GLM
Spiking	Yes	Partial	No	No
Neuron count	7,369	∼100	Whole brain	∼80K
Connectome	MPIN partial	Hand-coded	Activity map	Activity map
Behavior	Closed-loop	Open-loop	Open-loop	Closed-loop
Active inference	Analytic EFE	No	No	No
Motor control	Mauthner+RS	No	No	Tail kinematics
Neuromodulation	4-axis scalar	DA only	No	No
Senses	Vision+LL	Vision	Whole brain	Vision

The v2 model is unique in combining spiking neural dynamics with closed-loop behavior, active inference, and named reticulospinal neurons. However, it sacrifices neural realism for behavioral breadth: models like Dunn et al. (2016) achieve much higher neuron counts through whole-brain calcium imaging constraints, while Randlett et al. (2015) provide atlas-registered activity maps that could directly inform future versions.

12.5 Final Assessment

The Virtual Zebrafish Brain v2 represents a principled attempt to bridge computational neuroscience theory (predictive coding, active inference, three-factor STDP) with systems neuroscience data (MapZebrain atlas, identified reticulospinal neurons, RGC types). Its 77/100 decision quality score demonstrates that the behavioral architecture is competent. However, the honest assessment is that the spiking neural network substrate is largely bypassed by analytic formulas for the critical computations (goal selection, motor planning, sensory interpretation). The path to genuine biological plausibility requires making the spiking dynamics *do the work*—replacing hand-coded formulas with emergent computation in neural populations constrained by connectome data.

Overall biological plausibility score: On a scale from pure engineering (0) to biophysically faithful (10), we rate the v2 model at approximately **4/10**: the architectural skeleton maps onto real zebrafish neuroanatomy, but the computational substance remains largely algorithmic rather than neural.

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